```
In [11]:  # Make dependent variable as a factor (categorical)
          data$target = as.factor(data$target)

In [12]:  # Splitting the dataset into test and train
          print("Train Test Split") # 70/30 Split
          dt = sort(sample(nrow(data), nrow(data)*.7))
          train<-data[dt,]
          val<-data[-dt,]

          [1] "Train Test Split"

In [13]:  # No. of rows in Train and Val Dataset
          nrow(train)
          nrow(val)

          212
          91
```

Figure 2: Partitioning of the dataset

Loading and Understanding the dataset

```
In [4]:  data = read.csv("/content/heart_health.csv")

In [5]:  print("Minimum resting blood pressure")
         min(data$trestbps)
         print("Maximum resting blood pressure")
         max(data$trestbps)
         print("Summary of Dataset")
         summary(data)

 [1] "Minimum resting blood pressure"
94
 [1] "Maximum resting blood pressure"
200
 [1] "Summary of Dataset"
      age             sex               cp            trestbps
 Min.   :29.00   Min.   :0.0000   Min.   :0.000   Min.   : 94.0
 1st Qu.:47.50   1st Qu.:0.0000   1st Qu.:0.000   1st Qu.:120.0
 Median :55.00   Median :1.0000   Median :1.000   Median :130.0
 Mean   :54.37   Mean   :0.6832   Mean   :0.967   Mean   :131.6
 3rd Qu.:61.00   3rd Qu.:1.0000   3rd Qu.:2.000   3rd Qu.:140.0
 Max.   :77.00   Max.   :1.0000   Max.   :3.000   Max.   :200.0
      chol            fbs             restecg           thalach
 Min.   :126.0   Min.   :0.0000   Min.   :0.0000   Min.   : 71.0
 1st Qu.:211.0   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:133.5
 Median :240.0   Median :0.0000   Median :1.0000   Median :153.0
 Mean   :246.3   Mean   :0.1485   Mean   :0.5281   Mean   :149.6
 3rd Qu.:274.5   3rd Qu.:0.0000   3rd Qu.:1.0000   3rd Qu.:166.0
 Max.   :564.0   Max.   :1.0000   Max.   :2.0000   Max.   :202.0
     exang           oldpeak          slope             ca
 Min.   :0.0000   Min.   :0.00   Min.   :0.000   Min.   :0.0000
 1st Qu.:0.0000   1st Qu.:0.00   1st Qu.:1.000   1st Qu.:0.0000
 Median :0.0000   Median :0.80   Median :1.000   Median :0.0000
 Mean   :0.3267   Mean   :1.04   Mean   :1.399   Mean   :0.7294
 3rd Qu.:1.0000   3rd Qu.:1.60   3rd Qu.:2.000   3rd Qu.:1.0000
 Max.   :1.0000   Max.   :6.20   Max.   :2.000   Max.   :4.0000
      thal           target
 Min.   :0.000   Min.   :0.0000
 1st Qu.:2.000   1st Qu.:0.0000
 Median :2.000   Median :1.0000
 Mean   :2.314   Mean   :0.5446
 3rd Qu.:3.000   3rd Qu.:1.0000
```

Figure 3: Outcome of dataset -- minimum vs maximum blood pressure

blood pressure and chest pain type.

- View the confusion matrix and importance of each predictor.

The required data is fetched from the patient through IoT based smart wearables and is processed using the R tool. We will be using R version 3.3.2 (2016-10-31) and RStudio version 1.2.1335 here.

Now, load all the necessary libraries:

```
library(rpart) #used for building
classification and regression trees.
library(rpart.plot)
library(RColorBrewer) # help you choose
sensible colour schemes for figures
library(rattle) # provides a collection
of utilities functions for a data
scientist. library(randomForest) #Used
to create and analyse random forests.
```

Figure 1 shows the installation and loading of all the necessary packages.

Next, load the dataset:

```
data = read.csv("heart_health.csv")
```

Make the dependent variable a factor (categorical):

```
data$target = as.factor(data$target)
```

Now split the dataset into 'test' and 'train':

```
print("Train Test Split") # 70/30 Split
dt = sort(sample(nrow(data),
nrow(data)*.7)) train<-data[dt,]
val<-data[-dt,] nrow(train) nrow(val)
```

Figure 2 represents the partitioning, i.e., the splitting of the dataset into train set and test set in a 70 to 30 ratio, respectively.

Figure 3 helps understand the dataset being considered for the health analysis case study.

To analyse the data in the dataset, type:

```
print("Minimum resting blood pressure")
min(data$trestbps)
```